RESEARCH REPORT

An Awareness of Modern Latest AI Models

Task 02 | AI Model Research & Comparative Analysis

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Prepared by	TY PUNLEU
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Models Covered	6 Frontier & Open-Source AI Models

Abstract

The artificial intelligence landscape of 2025-2026 is defined by two structural shifts: first, the convergence of frontier model performance across closed-source and open-source providers, and second is the emergence of specialised architectures that outperform general-purpose models on specific task domains. This report examine and explore latest AI models, spanning large language modles (LLMs), multimodal systems, and open-weight alternatives.

Keywords: large language models, multimodal AI, open-source LLM, benchmark comparison, AI agent development, Gemini, GPT-5, Claude, DeepSeek, Llama, Grok

1. Introduction

The 2026 AI frontier is no longer a competition between just one or two dominant models. Since the release of GPT-4 in 2023, the rapid growth of both closed-source commercial lab (OpenAI, Anthropic, Google DeepMind, xAI) and open-souce (Meta, DeepSeek, Alibaba Qwen, and Google's own Gemma open-weight series).

Benchmark definitions used throughout this report are provided in Appendix A. All benchmark scores are sourced from independent analyses and official provider documentation published in 2025–2026.

2. Evaluation Criteria

Each model in this report is evaluated against the following standardised criteria:

- Model Name / Company: The official model designation and the lab that developed it.
- Model Type: Classfication as LLM, multimodal, open-source, reasoning model, or specialist system.
- Key Technical Specifications: Context window size, benchmark scores (GPQA, SWE-bench, ARC-AGI-2), pricing, and architecture (Mixture-of-Expert vs dense transformer).
- Comparison: A direct peer comparison with another model, assessing relative strengths, weaknesses, and optimal use cases.

3. Model 1 : GPT-5.4 (OpenAI)

Company	OpenAI (USA)
Release	August 2025; iterating through 2026 as GPT-5.x
Model Type	Multimodal LLM: text, image, audio, computer use; adaptive routing architecture
Context Window	Large (exact size tiered by plan; Thinking mode for complex tasks)
Coding Benchmark	74.9% SWE-bench Verified [1]
Reasoning Benchmark	92.8% GPQA [1]
API Pricing	$2.50 input / $15.00 output per 1M tokens [2]
Open Source	No, proprietary closed model

3.1 Description

GPT-5.4 is the current iteration of OpenAI's flagship model family, released as part of a strategic pivot from single-model releases to a tiered model family (GPT-5, GPT-5 mini, GPT-5 nano) OpenAI has positioned GPT-5 as an AI platform with the broades tool exosystem of any provider, including Microsoft Azure, Github Copilit, and the ChatGPT consumer product used by over 200 million weekly actice users.

Architecturally, GPT-5 is described as a "unified system" that employs an internal router to dynamically direct incoming prompts to the appropriate sub-model. As Pluralsight's 2026 AI report describes it: "a quick question might go to a fast 'main' model, while a complex problem is escalated to a deeper 'thinking' model" . This adaptive compute approach avoids the cost

inefficiency of always-on chain-of-thought reasoning while still providing access to extended reasoning when warranted.

3.2 Comparison vs. Claude Opus 4.6

On raw coding benchmarks, GPT-5.4 scores 74.9% on SWE-bench Verified, marginally behind Claude Opus 4.6 (80.8%) and Grok 4 (75%) . However, GPT-5.4 leads on agentic terminal execution via its Codex variant, scoring 77.3% on Terminal-Bench 2.0, a benchmark measuring CLI automation and DevOps scripting tasks . For natural prose and long-form written output, independent evaluations consistently place Claude ahead, with Claude capable of producing up to 128,000 output tokens in a single pass . GPT-5.4's primary competitive advantage is ecosystem breadth: it remains the most widely integrated AI model across enterprise software tools.

> *"GPT-5.4 is the best all-rounder with the largest ecosystem. Claude leads on natural prose and SWE-bench Verified. For agentic terminal tasks, GPT-5.4's Codex variant is the clear leader."* **- AI Models in 2026: Which One Should You Actually Use? (gurusup.com) [1]**

4. Model 2 : Claude Opus 4.6 (Anthropic)

Company	Anthropic (USA) - AI Safety focused lab, founded 2021
Release	February 2026 (beta); Claude 4.5 Sonnet available in production
Model Type	Hybrid reasoning LLM, safety-first architecture, extended thinking mode
Context Window	1,000,000 tokens (1M) [6]
Coding Benchmark	80.8% SWE-bench Verified, highest among all frontier models [5]
Reasoning Benchmark	91.3% GPQA [1]
API Pricing	$15.00 input / $75.00 output per 1M tokens (Opus); $3/$15 (Sonnet 4.5) [2]
Open Source	No, proprietary closed model

4.1 Description

Anthropic's Claude Opus 4.6 represents the current frontier of the Claude model family. The defining architectural feature of the 4.6 release is the extension of the context window to one million tokens, enabling the model to process an entire software repository, legal contract corpus, or multi-year financial dataset in a single inference pass .

Anthropic's differentiation philosophy is "Constitutional AI" (CAI), a training methodology where the model is taught to evaluate and correct its own outputs against a written set of principles, reducing the probability of harmful or deceptive outputs . This approach has attracted enterprise clients including financial institutions (NBIM, IG Group) and security companies (HackerOne, Palo Alto Networks) that require more predictable and auditable AI behaviour .

In practical developer tooling, Claude is the model powering the two most widely adopted AI coding editors: Cursor and Windsurf. This dominance in developer tooling is a meaningful market signal of Claude's real-world coding reliability.

4.2 Comparison vs. Gemini 3.1 Pro

On the SWE-bench Verified benchmark, which tests ability to resolve real, historical GitHub issues, Claude Opus 4.6 leads at 80.8% versus Gemini 3.1 Pro's 63.8% . On Terminal-Bench coding execution tasks, Claude also leads Gemini. However, Gemini 3.1 Pro leads on abstract reasoning: 94.3% GPQA versus Claude's 91.3%, and 77.1% on ARC-AGI-2 versus Claude's 68.8%. Cost is a significant differentiator: Gemini 3.1 Pro at $2/$12 per million tokens is substantially cheaper than Claude Opus 4.6 at $15/$75 .

5. Model 3 : Gemini 3.1 Pro (Google DeepMind)

Company	Google DeepMind (USA/UK)
Release	Gemini 3 launched late 2025; 3.1 Pro iteration in Q1 2026
Model Type	Multimodal LLM, text, image, audio, video; Mixture-of-Experts (MoE); three-tier reasoning
Context Window	1,000,000 tokens (1M) [8]
Coding Benchmark	63.8% SWE-bench [1]
Reasoning Benchmark	94.3% GPQA, leads all frontier models [1]
API Pricing	$2.00 input / $12.00 output per 1M tokens, most competitive frontier pricing [2]
Open Source	No, proprietary; Gemma series released open-source separately

5.1 Description

Gemini 3.1 Pro is Google DeepMind's current flagship model and the primary model available through the Google Agent Development Kit (ADK). Built on a sparse Mixture-of-Experts (MoE) architecture, Gemini dynamically activates only the subset of model weights relevant to each specific input, delivering more efficient inference than dense transformer alternatives .

The standout feature of the Gemini 3.x series is its "thinking model" capability: the model can allocate additional compute to reason through complex problems before producing a final output, similar to OpenAI's chain-of-thought reasoning but implemented as a first-class architectural feature rather than a prompting technique . This makes Gemini 3.1 Pro particularly effective for mathematical reasoning, multi-step logical inference, and structured planning tasks.

Gemini is also the only frontier model with native video understanding as part of its multimodal stack, in addition to text, image, and audio. Its 1M token context window is matched only by Claude Opus 4.6 among commercial closed-source models .

5.2 Comparison vs. GPT-5.4

On pure reasoning benchmarks, Gemini 3.1 Pro leads GPT-5.4: 94.3% vs. 92.8% GPQA and 77.1% vs. 52.9% on ARC-AGI-2 . On coding and agentic tasks, GPT-5.4 retains an advantage, particularly through its Codex-variant terminal execution capabilities. On multimodal image generation following specific instructions, ChatGPT/GPT-5.4 is consistently rated higher by independent evaluators . Pricing is a decisive differentiator: Gemini's $2/$12 per million tokens versus GPT-5.4's $2.50/$15 represents a meaningful cost saving at the production scale of a growing startup like CheckinMe . For ADK-based agent workloads, where tool-calling, session management, and response speed matter more than raw image generation quality, Gemini 3.1 Pro is the recommended default.

6. Model 4 : Grok 4 (xAI)

Company	xAI / Elon Musk (USA), integrated with X (Twitter) platform
Release	Mid-2025
Model Type	LLM with real-time internet and X/Twitter data integration; uncensored reasoning mode
Context Window	Large (exact size not publicly disclosed by xAI)
Coding Benchmark	75% SWE-bench, leads raw coding benchmarks [1]

Reasoning Benchmark	Competitive (exact GPQA not publicly published) [1]
API Pricing	$2.00 input / $15.00 output per 1M tokens (Grok 4.1: $0.20/$0.50) [2]
Open Source	No, closed source; Grok-1 weights released as historical reference only

6.1 Description

Grok 4 is xAI's flagship model and the only frontier AI system with live access to the X (formerly Twitter) data stream as a first-class feature, enabling real-time social intelligence, trend monitoring, and breaking-news analysis that static-knowledge models cannot replicate . xAI releases updates as continuous cloud-side improvements rather than major versioned releases, meaning Grok 4 in April 2026 is meaningfully more capable than Grok 4 at launch, without a version number change.

On raw SWE-bench coding benchmarks, Grok 4 leads all frontier models at 75%, though this lead is statistically marginal given that GPT-5.4 scores 74.9% and the benchmark's confidence intervals overlap . Grok's personality is intentionally less restricted than Claude or GPT-5.4, producing outputs that are more direct and less cautious, an advantage in contexts requiring unfiltered analysis, but a risk in regulated enterprise environments.

6.2 Comparison vs. GPT-5.4

On raw SWE-bench scores, Grok 4 (75%) and GPT-5.4 (74.9%) are effectively tied . GPT-5.4 has a substantially larger ecosystem, more stable enterprise pricing and SLAs, superior multimodal image generation, and broader integration with business tools. Grok's differentiators are its real-time X data access and its higher output per dollar on the budget Grok 4.1 API tier ($0.20/$0.50) . For CheckinMe's HR assistant use case, GPT-5.4 or Claude are more appropriate: both offer stronger safety controls and enterprise compliance frameworks. Grok 4 is most compelling for media analysis, social monitoring, and content moderation workflows.

7. Model 5 : DeepSeek V3.1 (DeepSeek AI)

Company	DeepSeek AI (China), subsidiary of High-Flyer Capital Management
Release	August 2025 (V3.1); January 2025 (V3 base and R1)
Model Type	Open-source LLM; Mixture-of-Experts; hybrid thinking/non-thinking modes; MIT License
Context Window	128,000 tokens [13]
Coding Benchmark	Surpasses V3 and R1 by 40%+ on SWE-bench and Terminal-bench [13]
Architecture	671B total parameters; ~37B active per token (MoE efficiency)
API Pricing	Free, open weights; self-host with no API cost; cloud inference via third parties
Open Source	Yes, MIT License; full model weights publicly available

7.1 Description

DeepSeek V3.1 is arguably the most geopolitically significant AI model of 2025. When DeepSeek V3 was released in January 2025, its training cost, reported at approximately USD 6 million, was compared against OpenAI's estimated USD 100 million for GPT-4, triggering a single-day USD 589 billion decline in Nvidia's market capitalisation as investors reassessed the assumption that frontier AI required frontier compute budgets .

DeepSeek V3.1 extends this foundation with a hybrid architecture supporting both deliberate "thinking" reasoning mode and fast "non-thinking" direct response mode, matching the design philosophy of OpenAI's adaptive routing and Gemini's thinking mode, but delivered as open weights that any organisation can download and run on their own hardware . All DeepSeek releases are published under the MIT License, meaning they are free for commercial use, modification, and redistribution without royalties or usage fees.

7.2 Comparison vs. Claude Opus 4.6

DeepSeek V3.1's most significant advantage over Claude Opus 4.6 is cost and deployment flexibility: it can be self-hosted on-premises for zero API cost, while Claude Opus 4.6 costs $15/$75 per million tokens with no self-hosting option.

On raw capability, Claude Opus 4.6 leads on SWE-bench Verified (80.8% vs. DeepSeek's competitive but not publicly confirmed score), benefits from Anthropic's Constitutional AI safety training, and carries enterprise SLA guarantees that open-source self-hosting cannot match . The practical conclusion is that DeepSeek V3.1 is appropriate for internal tooling and research workloads where data sovereignty matters; Claude Opus 4.6 is appropriate for customer-facing HR assistant features where output quality and safety are paramount.

8. Model 6 : Meta Llama 4 (Meta AI)

Company	Meta AI (USA)
Release	April 2025
Model Type	Open-source multimodal LLM; Mixture-of-Experts; native multilingual; natively multimodal
Context Window	Up to 10,000,000 tokens (10M), largest available context window [15]
Languages Supported	200+ languages in training; fine-tuned for 12 including Arabic, Hindi, Spanish, German
Architecture	Maverick: 17B active / 400B total (128 experts). Scout: 17B active / 109B total (16 experts) [15]
API Pricing	Free, open weights; inference via Meta, Together AI, Fireworks, or self-hosted

Open Source	Yes, open weights under Meta's commercial license

8.1 Description

Meta Llama 4 is the most downloaded open-source AI model family in 2025, inheriting the massive community ecosystem built around Llama 3. The defining architectural innovation in Llama 4 is early fusion multimodality: unlike previous models that added vision capabilities as a separate component after text pretraining, Llama 4 integrates image and text understanding from the beginning of pretraining . This produces more coherent image-text reasoning than "bolt-on" vision systems.

Llama 4's 10-million token context window, larger than any other available model, is the result of Meta's research into efficient attention mechanisms. In practice, retrieval quality at extreme context lengths is still an active research area, but for document-heavy use cases like legal review or HR policy analysis, even partial utilisation of this window is practically useful .

8.2 Comparison vs. DeepSeek V3.1

Both Llama 4 and DeepSeek V3.1 are open-source MoE models, but they target different strengths. Llama 4 has the broader community support (more fine-tuned variants, more deployment tooling, more documentation) and stronger multilingual coverage, critical for applications serving non-English markets including Khmer. DeepSeek V3.1 has stronger chain-of-thought reasoning performance and is more optimised for technical coding tasks.

9. Consolidated Comparison Table

Table 1 presents a side-by-side summary of all nine models across key evaluation dimensions. Gemma 4 (Model 9) was released April 2, 2026 and is included as an addendum based on official Google DeepMind documentation and independent day-one analysis.

Table 1: Frontier and Open-Source AI Model Comparison (April 2026)

Model	Company	Type	Context	SWE-bench	GPQA	$/1M tok (in/out)	Open Source
GPT-5.4	OpenAI	Multimodal LLM	Large	74.9%	92.8%	$2.50 / $15.00	No
Claude Opus 4.6	Anthropic	Hybrid Reasoning LLM	1M	80.8%	91.3%	$15.00 / $75.00	No
Gemini 3.1 Pro	Google	Multimodal MoE LLM	1M	63.8%	94.3%	$2.00 / $12.00	No
Grok 4	xAI	LLM + Real-Time Web	Large	75.0%	N/A	$2.00 / $15.00	No
DeepSeek V3.1	DeepSeek	Open MoE LLM	128K	40%+ vs V3	Comp.	Free / Self-Host	Yes (MIT)
Meta Llama 4	Meta	Open Multimodal MoE	10M	Good	Good	Free / Self-Host	Yes
Mistral Large	Mistral AI	Open/API MoE LLM	128K	Solid	Solid	~$2 / $6	Partial
DeepSeek V4	DeepSeek	Open Multimodal MoE+Mem	1M	~80%+ (est)	TBD	~$0.30 (est.)	Yes (Apache 2.0)
Gemma 4 (31B/26B MoE)	Google DeepMind	Open Multimodal LLM	256K	80.0% LCBv6	82.3% GPQA-Dia.	Free / Self-Host	Yes (Apache 2.0)

Notes: Benchmarks sourced from independent analyses and provider documentation (2025–2026). SWE-bench Verified tests ability to resolve real GitHub issues. GPQA tests graduate-level scientific reasoning. ARC-AGI-2 tests abstract reasoning. Costs are API input/output per million tokens and subject to change. 'N/A' = not publicly disclosed. 'Est.' = estimated from pre-release information. Gemma 4 GPQA score uses GPQA Diamond variant [20]. LCBv6 = LiveCodeBench v6. Gemma 4 released April 2, 2026 [19].

Table 2: Key Benchmark Definitions

Benchmark	Full Name	What It Measures	Best Score (Apr 2026)
SWE-bench Verified	Software Engineering Bench.	Resolving real GitHub issues from open-source repos	Claude 4.6: 80.8%
GPQA	Grad-level Google-Proof Q&A	Expert-level scientific reasoning (PhD-level questions)	Gemini 3.1: 94.3%
ARC-AGI-2	Abstraction & Reasoning Corpus v2	Abstract pattern recognition and novel reasoning	Gemini 3.1: 77.1%
Terminal-Bench	Terminal Execution Bench.	CLI automation, DevOps scripting, agentic execution	GPT-5.3-Codex: 77.3%
HumanEval	Human Evaluation	Standard code generation correctness across problems	DeepSeek V4 est.: 90%

10. Conclusion

The era of a single dominant model has passed. GPT-5.4 leads on ecosystem integration, Claude Opus 4.6 on coding accuracy and safety, Gemini 3.1 Pro on reasoning and cost efficiency, Grok 4 on real-time data, DeepSeek V3.1 on open-source cost efficiency, Llama 4 on multilingual reach, Mistral on European compliance and instruction precision, DeepSeek V4 on next-generation architecture, and Gemma 4 on open-model on-device intelligence.

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Appendix A : Benchmark Definitions

The following benchmark definitions are referenced throughout this report.

A.1 SWE-bench Verified

SWE-bench Verified (Software Engineering Benchmark Verified) is an industry-standard benchmark that tests an AI model's ability to resolve real, historical software bugs and feature requests submitted as GitHub Issues against open-source Python repositories. Unlike synthetic coding benchmarks, SWE-bench uses actual production issues that required human developer judgment to resolve, making it the most realistic proxy for real-world software engineering capability. Scores represent the percentage of issues resolved correctly without human intervention. Current leader: Claude Opus 4.6 at 80.8% [5].

A.2 GPQA (Graduate-Level Google-Proof Q&A)

GPQA tests a model's ability to answer PhD-level questions in biology, chemistry, and physics, questions specifically designed to be unanswerable by internet search alone, requiring genuine domain reasoning. The "Google-proof" constraint means that scores reflect genuine reasoning capability rather than retrieval from memorised training data. Current leader: Gemini 3.1 Pro at 94.3% [1].

A.3 ARC-AGI-2 (Abstraction and Reasoning Corpus v2)

ARC-AGI-2 is the second version of François Chollet's Abstraction and Reasoning Corpus, designed to test core fluid intelligence, the ability to reason about novel patterns not seen in training data. ARC-AGI-2 is significantly harder than the original ARC and is considered one of the most reliable proxies for general reasoning capability. Current leader: Gemini 3.1 Pro at 77.1%, more than doubling predecessor scores [1].

A.4 Terminal-Bench

Terminal-Bench is a benchmark that evaluates a model's ability to perform agentic tasks using a live Unix terminal, including complex system administration, CLI automation, environment configuration, and DevOps scripting. Unlike SWE-bench which tests code generation, Terminal-Bench tests multi-step execution in a real system environment. Current leader: GPT-5.3-Codex at 77.3% [5].

A.5 HumanEval

HumanEval is OpenAI's original code generation benchmark consisting of 164 handwritten Python programming problems. Models are scored on pass@1, the percentage of problems solved correctly on the first attempt. While now considered a baseline benchmark (most frontier models score above 85%), it remains useful for assessing code generation correctness on well-defined function-level tasks.

Appendix B : AI Model Release Timeline (2025–2026)

Figure 1 below summarises the release chronology of models covered in this report.

Period	Model Released	Provider	Significance
Jan 2025	DeepSeek V3 + R1	DeepSeek	Open-source frontier; $6M training cost shocks market; Nvidia stock drops 18%
Apr 2025	Meta Llama 4	Meta	Open multimodal MoE; 200+ languages; 10M context window
Aug 2025	DeepSeek V3.1	DeepSeek	Hybrid thinking/non-thinking mode; 40%+ improvement on SWE-bench
Aug 2025	GPT-5 family	OpenAI	Tiered model family; adaptive routing; platform strategy declared
Late 2025	Gemini 3 / 3.1 Pro	Google	MoE reasoning leader; ARC-AGI-2 best; ADK integration
Late 2025	Grok 4	xAI	Real-time X data; leads raw SWE-bench; uncensored reasoning
Feb 2026	Claude Opus 4.6 (beta)	Anthropic	1M context; leads SWE-bench Verified at 80.8%; safety focus
Apr 2, 2026	Gemma 4 (4 sizes)	Google DeepMind	Released April 2, 2026; E2B/E4B edge + 26B MoE + 31B Dense; Apache 2.0; native multimodal; on-device to cloud
Q2 2026 (est.)	DeepSeek V4	DeepSeek	1T param MoE; Engram memory; Huawei Ascend hardware; Apache 2.0
Q2 2026 (est.)	GPT-5.5 (Spud)	OpenAI	Pre-training complete; imminent release per Sam Altman
Q2 2026 (est.)	Grok 5	xAI	Q2 target per xAI internal projections

— End of Report —

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